

Coursework 1: Beat Tracking System *

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1 Introduction

The aesthetic dimensions of music are a subject of ongoing exploration, and there is growing evidence that they are amenable to systematic analysis. As Handel [1] noted, the experience of rhythm involves a complex interplay of movement, regularity, grouping, accentuation, and differentiation. These elements contribute to the overall structure of a musical composition, and they have been studied extensively by researchers in fields such as musicology, psychology, and neuroscience. By analyzing the patterns of rhythm in musical pieces, researchers can gain insights into the cultural and historical contexts in which they were created, as well as the emotional and psychological effects that they produce in listeners.

As Eric D. Scheirer [2] pointed out, establishing a ground truth for rhythm requires the consensus of human listeners on the rhythmic aspects of the musical content. Therefore, focusing on a specific style of music, as Dixon et al. [3] did, can be a reasonable approach to adjust the beat tracking method and improve its accuracy.

By limiting the scope of the study to a particular genre, the variations in rhythm that exist across different genres can be minimized, allowing for a more accurate and focused analysis. Therefore, focusing on a particular style of music can be a valid and practical strategy to address the challenges in beat tracking, and the results can have implications not only for the specific genre but also for the general field of music analysis.

In this report, I also focus specifically on the rhythmic aspects of ballroom dance music and explore an approach for characterizing them that can be used for music retrieval tasks. Considering the requirements of the coursework, the method in this report will be based on some existing algorithms and make innovative adjustments.

The data set is from ISMIR2004¹. The annotations used are provided by Florian Krebs².

2 Background

The term “onset detection” refers to the identification of the starting points of distinct events in acoustic signals. A perception of an onset is triggered by a noticeable alteration in the loudness, pitch, or timbre of the sound [4].

Onset detection encompasses a wider range of applications, including not only music but also a variety of acoustic events, such as the beginning of a spoken word or the animal sounds in a natural soundscape. Moreover, beat-tracking or the detection of rhythmic patterns in music is always closely related to onset detection.

Dixon et al. [3] proposed an innovative approach to characterize musical pieces by extracting dominant bar-length rhythmic patterns. Their hypothesis suggests that rhythmic patterns are not randomly distributed among musical genres and can indicate a specific genre or a small set of possible

genres. To achieve genre classification, they proposed an automatic extraction technique that identifies the most prominent pattern for each piece in a music database of Standard and Latin ballroom dance music. This technique is a significant step forward in creating musically relevant, higher-level parameters that can be used in genre classification and music retrieval tasks.

However, in the field of tempo and beat analysis, it is important to note that many methods require onset detection as a precursor to beat analysis. As Scheirer et al. [2] pointed out, “robust onset detection precedes beat analysis, which entails an important restriction to their applicability.” The limitation arises from the fact that not all music genres have clear and distinct onsets, making it difficult to apply these methods to a diverse range of musical styles.

Therefore, beat tracking methods require a trade-off between accuracy and applicability. While more complex and accurate algorithms may work well for a limited range of musical styles or specific instrumentation, simpler and more general algorithms may sacrifice accuracy for wider applicability. This trade-off is an important consideration when selecting a beat tracking method for a particular application.

3 Description of Algorithms

3.1 Time Domain: RMS

The RMS energy method assumes that the onset of an audio signal is always accompanied by a sudden increase in the signal’s amplitude level, which is quite reasonable and understandable, but it seems that this method has limited accuracy.

3.2 Frequency Domain: Spectral Flux

Magnitude-based onset detection methods such as SF and HFC (High Frequency Content) are limited in that they only use energy, power, or magnitude information. However, they are effective for detecting percussive onsets due to the sharp, transient nature of percussive sounds. By comparing the spectral magnitudes of successive frames of an audio signal, the SF method detects sudden changes in spectral content that correspond to onsets.

The dataset used in this coursework is particularly suitable for onset detection methods that rely solely on magnitude information. According to an Empirical Comparison table provided by Dixon [5], the SF (Spectral Flux) and RCD (Rectified Complex Domain) methods perform the best in the pitched percussive (PP) and non-pitched percussive (NP) categories. As such, I chose to use the SF method for onset detection in the ballroom dataset, which is composed of music with strong rhythmic patterns.

The definition of the spectral flux formula, as stated by Dixon [5], is:

$$SF(n) = \sum_{k=-\frac{N}{2}}^{\frac{N}{2}-1} H(|X(n, k)| - |X(n-1, k)|)$$

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¹<http://mtg.upf.edu/ismir2004/contest/tempoContest/data1.tar.gz>

²<https://github.com/CPJKU/BallroomAnnotations>

where n is the frame index, k is the frequency bin index, N is the FFT size, $X(n, k)$ is the magnitude of the STFT at frame n and frequency bin k , and $H(\cdot)$ is the half-wave rectifier function.

Next, I will explain how I built this onset detection function in the code.

After obtaining the necessary information, such as the sampling rate, from the input file, the audio signal is zero-padded to ensure that the first frame is centered at time 0. The linear magnitude values are obtained by calculating the absolute value of the STFT using a Hamming window function, which is computed in one go to avoid redundant calculations that may decrease algorithm performance. The spectral flux value is then calculated for each frame of the STFT and stored in a one-dimensional array. The resulting ODF values are then normalized to scale between 0 and 1, ensuring that the highest peak in the ODF has a value of 1. The generated ODF is returned as the output of the function. This method is particularly effective for detecting percussive onsets in audio signals.

4 Evaluation

4.1 F-measure Method

The F-measure method is widely used in the evaluation of information retrieval (IR) systems. The values of recall R and the values of precision P are all collected for the final evaluation. The F-measure of the system is defined as the weighted harmonic mean of its precision and recall [6] :

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}}$$

where the weighted value $\alpha \in [0, 1]$.

The calculated F-measure is only high when both R and P are high. Therefore, scholars generally believe that F-measure is a comprehensive evaluation standard that considers multiple aspects at the same time.

4.2 Accuracy Calculation

The subset `ballroom-subset` is manually selected from the whole data set, and arrange the `.wav` files and `.beats` files according to the same structure of the original data set `ballroom` and corresponding annotation files.

Accuracy can be calculated by comparing the predicted value with the Ground Truth value. The threshold is set to 0.05, which means only 5% error is allowed. The value compared is BPM.

Table 1 shows the printed prediction values and their corresponding ground truth, and the final result is 18.75% accuracy.

This also shows that the current algorithm still has a lot of room for improvement. For example, do an Envelope extraction before onset detection.

5 Reflection

In the supporting code provided for this coursework, certain aspects are lacking in their integrity, particularly the Evaluation method, which is not rigorous enough. By comparing the value of F-measure, more meaningful conclusions can be drawn.

Furthermore, it should be noted that the initial data preprocessing in the code could be more concise. This lack of

Estimated	Ground Truth
93.8173469387755	125.00000000000001
121.10736196319019	122.44897959183669
144.5644171779141	124.99999999999999
119.95051546391753	122.44897959183675
125.97216494845362	124.99999999999999
98.46932515337424	125.00000000000001
157.125	117.64705882352939
113.14591836734694	124.99999999999999
0.12738853503184713	99.99999999999999
139.85348360655738	101.69491525423727
87.04563894523326	101.69491525423727
119.18431771894093	98.84678747940691
132.94207317073167	98.36065573770492
147.334008097166	94.9367088607595
118.91463414634146	96.77419354838725
120.34451219512196	101.69491525423739

Table 1: Evaluation Results

concision has resulted in a series of difficulties that arose during the course of the project. As such, a complete rewrite of the preprocessing step may lead to a more streamlined process and improve the overall efficiency of the algorithm.

The final evaluation accuracy rate of less than 20% highlights the need for further development of the algorithm in detecting ballroom music. As such, it is recommended that beat tracking be expanded to consider multiple features of acoustic sound such as pitch, harmonics, energy level, zero-crossing rate, etc. Incorporating these attributes and weighting them differently could improve accuracy, even if only energy, power, or magnitude is detected. Additionally, the introduction of contextual information into the beat-tracking system would benefit the system.

Finally, to further improve the robustness and reliability of the algorithm, it would be interesting and meaningful to compare the results obtained using SF with those obtained using ‘Superflux’ in addition to the SF algorithm. This comparison would allow for a more thorough understanding of the algorithm’s performance and provide insight into potential areas for future improvement.

References

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